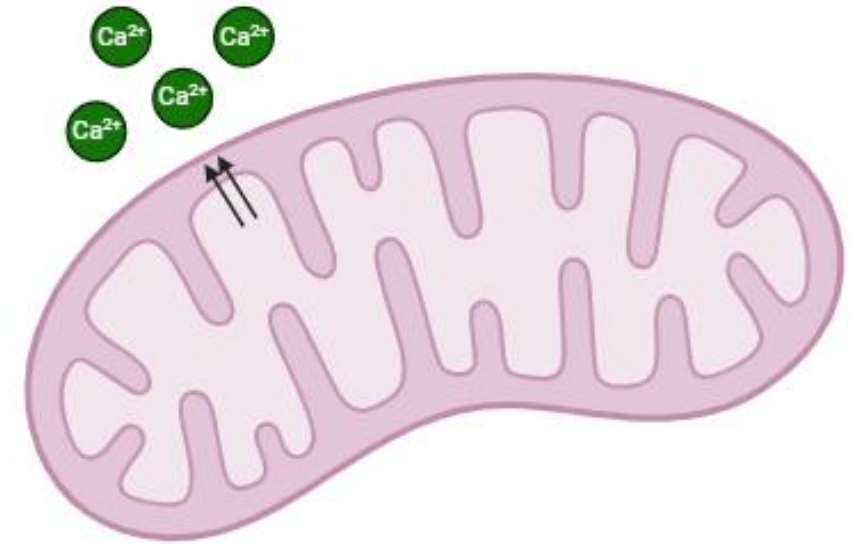
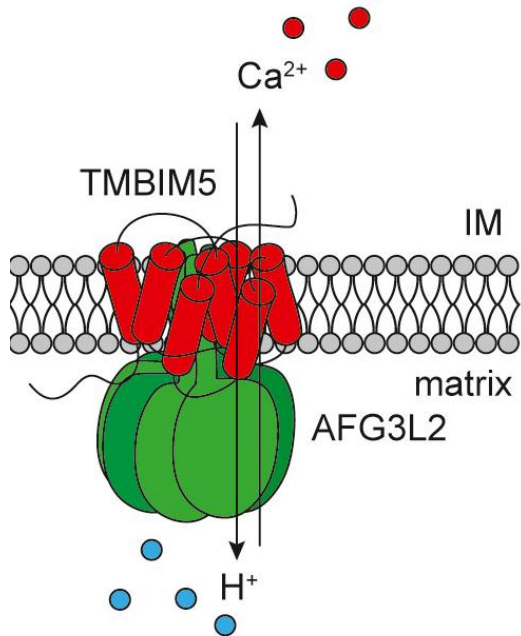


TMBIM5 as a $\text{Ca}^{2+}/\text{H}^{+}$ exchanger in the mitochondrial inner membrane



Mitochondrion

TMBIM5(MICS1)

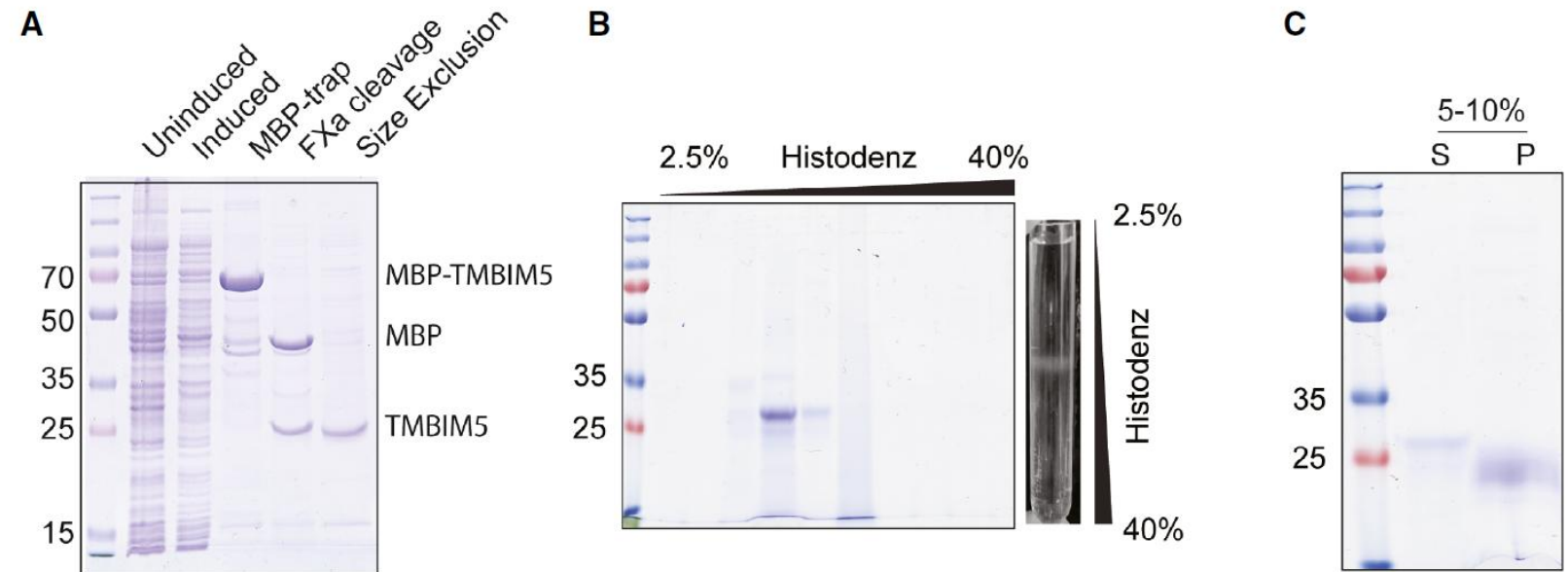


Question:

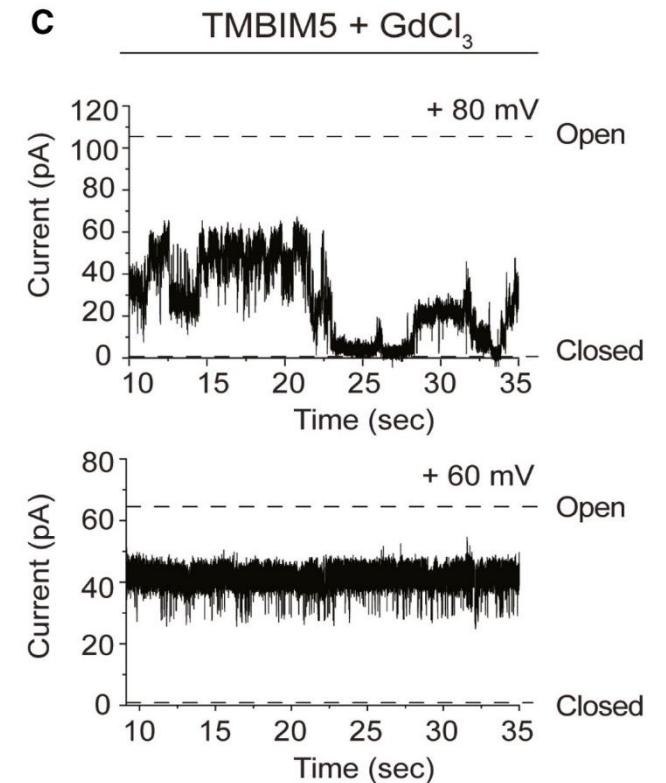
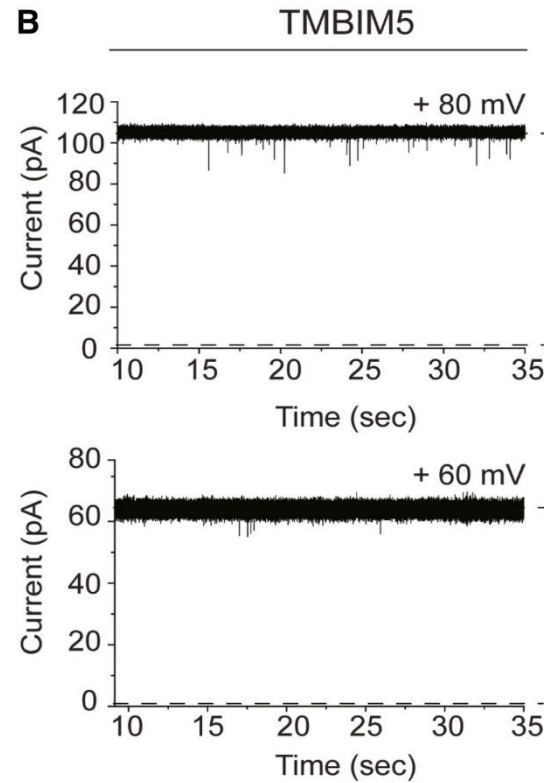
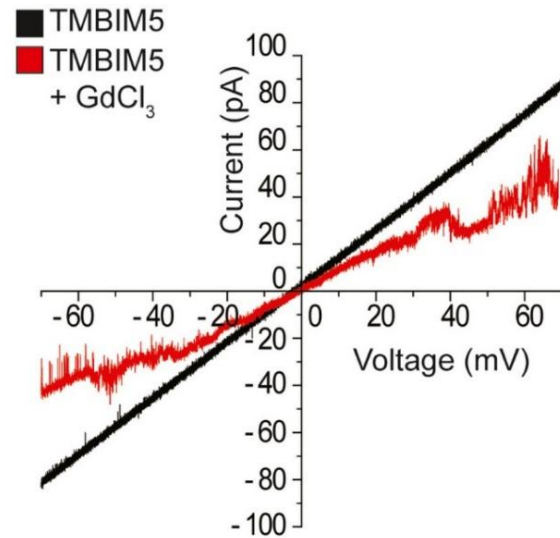
Does TMBIM5 forms a calcium ion channel ?

Aim and approach:

1. Purification of TMBIM5
2. Reconstitution of TMBIM5 into LUVs and perform electrophysiological characterization
3. Perform Gadolinium chloride sensitivity test to confirm it's a calcium channel

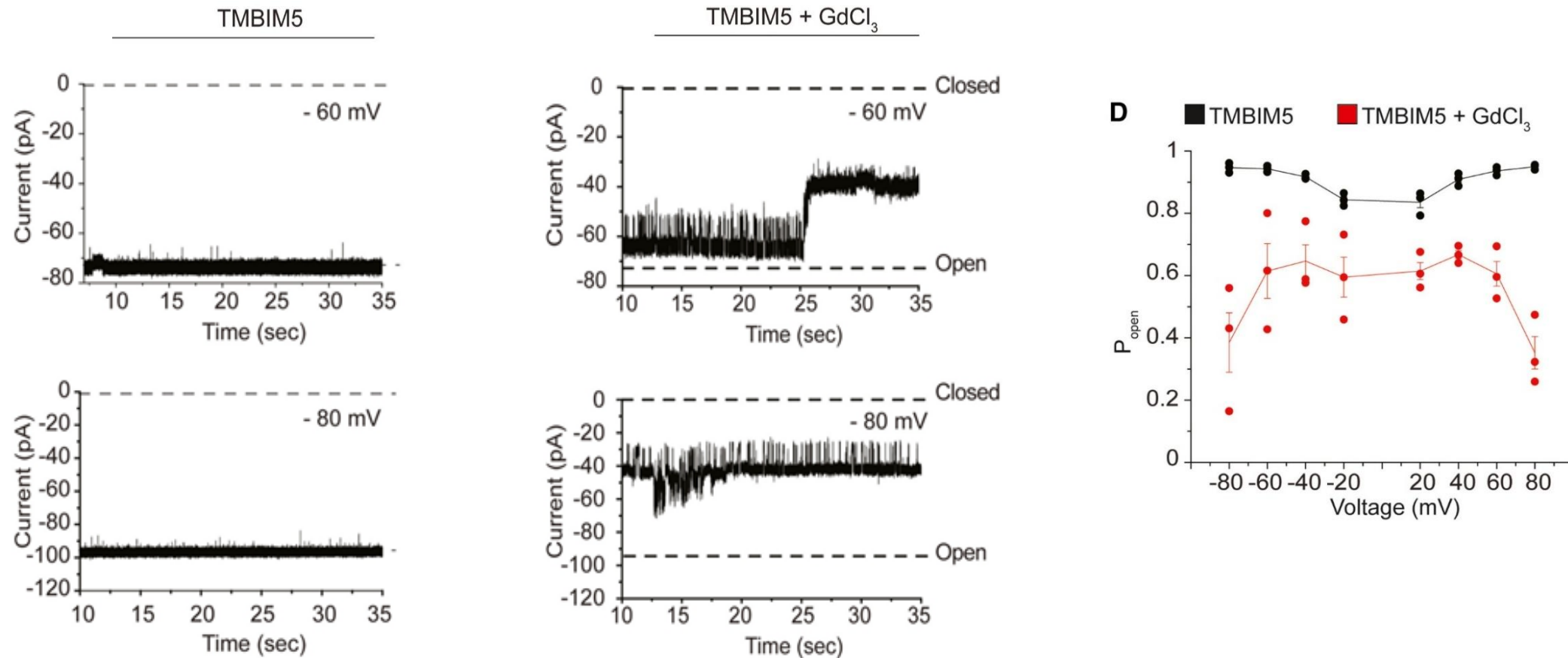


Reconstitution of TMBIM5 activity in lipid bilayers



Patron et al., EMBO J. 2022

Reconstitution of TM BIM5 activity in lipid bilayers



Patron et al., EMBO J. 2022

TM BIM5 as a mitochondrial Ca²⁺ ion channel in the mitochondrial inner membrane